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Module 13911 - Algebra: Structures and Algorithms - Lecture

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0 The Very Basics

05.10.2026

The key structures in algebra: groups, rings and fields.

Definition (Abelian) Group

An *abelian group* is a set G with an operation $* : G \times G \to G$ which satisfies the following $\forall a, b, c \in G$:

$$a * b = b * a \tag{C}$$
$$a * (b * c) = (a * b) * c \tag{A}$$
$$a * e = a \tag{N}$$
$$a * a^{-1} = e \tag{I}$$

$\Rightarrow$ without the commutativity (C) G would just be called a *group*.

$\rightarrow$ Group Theory (Abelian Groups are much easier than general ones)

Definition Ring

A *ring* is a set R with the operations $+ : R \times R \to R$ and $\cdot : R \times R \to R$ which satisfies the following $\forall a, b, c \in R$:

CA : $a + b = b + a, \forall a, b \in R$
AA : $a + (b + c) = (a + b) + c, \forall a, b, c \in R$
NA : There exists a unique element in R, denoted by 0 or 0_R, such that $a + 0 = a, \forall a \in R$
IA : For every $a \in R$ there exists a unique element $-a \in R$ such that $a + (-a) = 0$
AM : $a \cdot (b \cdot c) = (a \cdot b) \cdot c, \forall a, b, c \in R$
D : $a \cdot (b + c) = a \cdot b + a \cdot c, \forall a, b, c \in R$ and $(a + b) \cdot c = a \cdot c + b \cdot c, \forall a, b, c \in R$

$\rightarrow$ A ring $(R, +, \cdot)$ is called *commutative* when:

CM : $a \cdot b = b \cdot a, \forall a, b \in R$

$\rightarrow$ A ring $(R, +, \cdot)$ is called a *unitary* when:

NM : There exists a unique element in R, denoted by 1 or 1_R, such that $a \cdot 1 = a, \forall a \in R$

$\rightarrow$ A ring $(R, +, \cdot)$ is called *proper* if in that ring one has $0_R \neq 1_R$.

$\rightarrow$ Theory of commutative rings is called commutative algebra.

Definition Field

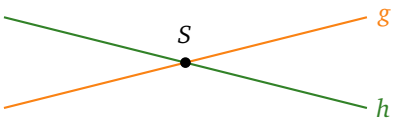
A proper, unitary, commutative ring $(R, +, \cdot)$ is called a *field* if all non-zero elements of R have a multiplicative inverse.

$\rightarrow$ Galois Theory, Theory of finite fields, algebraic closure, etc.

1 Algebra, Geometry and Algorithms

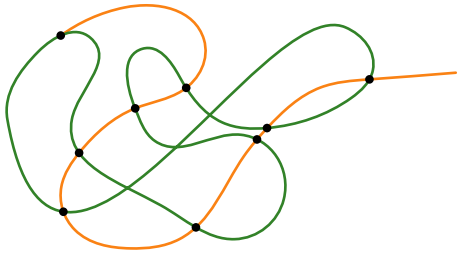
Linear Algebra

- Algebra: linear equations
- Geometry: sets defined by linear subsystems of equtions, affine subspaces, vectors, vector subspaces, ...
- Algorithms: Gaussian elimination



Non-Linear Algebra

- Algebra: polynomial equations
- Geometry: Varieties (sets defined by systems of polynomial equations)
- Algorithms: Buchberger's algorithm (a far-reading generalization of Gaussian elimination), Gröbner bases



The more Technical Layer

Notation of ideal in a ring
→ this notation is relevant for the above topics but also for construction of rings and fields
(the latter is used in coding theory and cryptography)

1.1 Polynomials and the Affine Space

Variables considered per default in algebra are *formal variables (symbols)*. Formal means roughly placeholder for anything. There is no domain attached to this.

Formal variables are used in *formal expressions (symbolic expressions)*.

Note the difference to functions: Symbolic expressions are just expressions, not functions. In particular, they don't have any domain or co-domain.

In what follows, we consider n formal variables ($n \in \mathbb{Z}_{>0}$) whose default names will be $x_1, \ldots, x_n$. For $n = 1$, it is customary to use x, If $n = 2$ it's cosumary to use x and y. For $n = 3$ it's cosumary to use x, y, z.

Definition 1.1.1

Let $x = (x_1, \ldots, x_n)$ be the vector of n formal variables and $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{Z}^n_{\geq 0}$ a vector of n non-negative integers. Then the expression

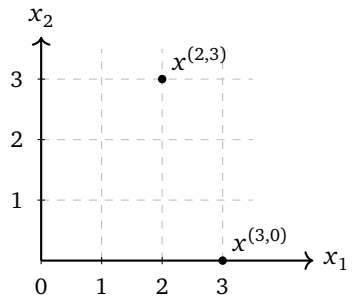
$$x^\alpha := x_1^{\alpha_1} \cdot \ldots \cdot x_n^{\alpha_n}$$

is called the *monomial for the expotent vector* α. $\alpha_1 + \ldots + \alpha_n$ is called the *(total) degree of the monomial* x^α, denoted as $|\alpha|$.

Our *coefficient domain* will be an arbitrary field which we denote by k thoughout.

Example

Given: $x^{(2,3)} = x_1^2 \cdot x_2^3$ and $x^{(3,0)} = x_1^3$



You already know some fields: $\mathbb{Q}, \mathbb{R}, \mathbb{CF}_p$ and $\mathbb{F}_{p^N}$ where p is prime, $\mathbb{Q}[\sqrt{2}]$

→ It's important to know some examples of them.

Definition 1.1.2

A *polynomial in variables* $x_1, \ldots, x_n$ with coefficients in k is a formal expression of the form

$$f = \sum_{\alpha} c_\alpha x^\alpha, \tag{1}$$

where α runs over $\mathbb{Z}^n_{\geq 0}$ and $c_\alpha \neq 0$ for any finitely many choices of k.

(You must have seen polynomials in one variable before... surprise, surprise - life is not one-dimensional!)

$c_\alpha x^\alpha$ (when $c_\alpha \neq 0$) is called a *term of degree* $|\alpha|$ of f. The *(total) degree of* f is the maximum degree of the terms in f:

$$\deg f = \max\{|\alpha| : \alpha \in \mathbb{Z}^n_{\geq 0}, c_\alpha \neq 0\}$$

The *zero polynomial* $f = 0$ has no terms and by this degree $-\infty$.
The *set of all polynomials in* $x_1, \ldots, x_n$ with coefficients in k is denoted by $k[x_1, \ldots, x_n]$.

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Note: Polynomials are formal expressions, not functions.

Equality of two polynomials is defined through comparison of coefficients.

We make $k[x_1, \ldots, x_n]$ an algebraic structure by introducing the basic operations, which are the addition and the multiplication.

Addition:

$$\sum_{\alpha} c_\alpha \cdot x^\alpha + \sum_{\alpha} d_\alpha \cdot x^\alpha := \sum_{\alpha} (c_\alpha + d_\alpha) \cdot x^\alpha$$

Multiplication:

$$\left(\sum_{\alpha} c_\alpha \cdot x^\alpha\right)\left(\sum_{\beta} d_\beta \cdot x^\beta\right) := \sum_{\alpha,\beta} c_\alpha \cdot d_\beta \cdot x^{\alpha+\beta} = \sum_{\gamma} \left(\sum_{\alpha,\beta:\alpha+\beta=\gamma} c_\alpha \cdot d_\beta\right) \cdot x^\gamma$$

Adrian's Question

Adrian asked about the role of the condition $c_\alpha \neq 0$ for finitely many α. It makes the number of terms finite and allows one to define evaluation of polynomials later on. But: if one remove the finiteness assumption, what one gets is the notion of a formal power series. The set of all formal power series with coefficients in k and variables $x_1, \ldots, x_n$ is denoted as $k[[x_1, \ldots, x_n]]$. They have applications in different situations, including combinatorics (generating functions).

Remark 1.1.3

We can view k as a subset of $k[x_1, \ldots, x_n]$ and by this $k[x_1, \ldots, x_n]$ contains $1 \in k$ (The multiplecative identity).
It turns out that $(k[x_1, \ldots, x_n], +, \cdot)$ is a unitary, commutative ring:

$$f + (h + g) = (f + h) + g \qquad f \cdot (h \cdot g) = (f \cdot h) \cdot g \qquad (f + g) \cdot h = f \cdot h + g \cdot h$$
$$f + g = g + f \qquad f \cdot g = g \cdot f$$
$$f + 0 = f \qquad f \cdot 1 = f$$
$$f + (-f) = 0$$

This holds for all $f, g, h \in k[x_1, \ldots, x_n]$. The $-f$ is the unique polynomial whose addition to f gives 0.

Remark 1.1.4

There are three basic choices of n:

n = 1	x_1	x
n = 2	x_1, x_2	x, y
n = 3	x_1, x_2, x_3	x, y, z

Later, a connection to geometry will be established. These three basic cases will get connected to line resp. plane and resp. 3-dim. space.

Definition 1.1.5

In algebraic geometry, the space k^n of all n-tuples with the coefficients in k is called the *n-dimentional affine space over the field k*.
From linear algebra we know, that k^n is a k-vector space.

Fun Fact: It's by far not the only space one can consider. Another very common and important space is the projective space. Actually, in the Elliptic Curve Cryptography (ECC), the elliptic curve is considered in the projective space.

Definition 1.1.6

We can evaluate $f = \sum_\alpha c_\alpha \cdot x^\alpha \in k[x_1, \ldots, x_n]$ at a point $(a1, \ldots, a_n) \in k^n$ of the affine space obtaining the value.

$$f(a_1, \ldots, a_n) = \sum_{\alpha=(\alpha_1,\ldots,\alpha_n)} c_\alpha a_1^{\alpha_1} \cdots a_n^{\alpha_n}$$

This gives rise to to a function $k^n \to k$. This function is usually denoted again by f, even though $f \in k[x_1, \ldots, x_n]$ and $f : k^n \to k$ is not the same thing. In math, such a situation is called abuse of notation.

I know it's not quite right, but I know what I'm doing

Functions arising this way are called *polynomial functions*.

Remark 1.1.7

In some fields, for example, in finite fields k, different polynomials in $k[x_1, \ldots, x_n]$ may define the same polynomial function.

Example

Consider the finite field $\mathbb{F}_2 = \{0,1\}$ with

$+$	0	1
0	0	1
1	1	0

and

$\cdot$	0	1
0	0	0
1	0	1

$f = x^2 + x \in \mathbb{F}_2[x]$
$g = x^3 + x \in \mathbb{F}_2[x]$

$f : \mathbb{F}_2 \to \mathbb{F}_2$

$a \in \mathbb{F}_2$	$f(a) \in \mathbb{F}_2$
0	0
0	0

$g : \mathbb{F}_2 \to \mathbb{F}_2$

$a \in \mathbb{F}_2$	$g(a) \in \mathbb{F}_2$
0	0
0	0

So, both functions are the same, but the polynomials are different

Fun Fact: Do we need $\mathbb{F}_2[x]$ for anything?
→ Yes! Advanced Encryption Standard (AES) used the field $\mathbb{F}_{256}$, whose construction requires taking a polynomial from $\mathbb{F}_2[x]$.

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Note: Test today → some will need to catch up
→ Check-In Date for the catch-up cohort: 9th of November, you show your progress on the form of the written solutions of the exercises from a textbook (linear algebra over arbitrary fields, with proofs).

Entry Level Exercise

1. Formulate mathematical induction (with words and/or formulas).
2. Define the notion of a group. Give an example of an abelian and a non-abelian group.
3. Define the notion of a ring. Give an example of a commutative and a non-commutative ring.
4. Define the notion of a field. Give an example of an infinite and finite field.
5. Let k be a infinite field and consider a univariate polynom $f \in k[x]$.
   Proof: $f = 0 \iff f(a) = 0, \forall a \in k$.

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Theorem 1.1.8

Let the field k be infinite. Then $f \in k[x_1, \ldots, x_n]$ is a zero polynomial (all coefficients of f are zero) if and only if $f : k^n \to k$ is a zero function.

Proof: The implikation ($\Rightarrow$) is trivial because $f = \sum_\alpha c_\alpha x^\alpha$ gets evaluated to zero on k^n when all of the c_α's are zero. So, the actual work is to show the converse ($\Leftarrow$). For the converse, we are going to use the mathematical induction on n.

We consider the case $n = 1$. as a basis of the induction. We consider a polynomial f in one variable x, which is <u>not</u> a zero polynomial and show that $f(a) \neq 0$ for some $a \in k$. We can write f as $f = \sum_{i=1}^{d} c_i x^i$ where $d \in \mathbb{N}_0$ and $c_d \neq 0$.

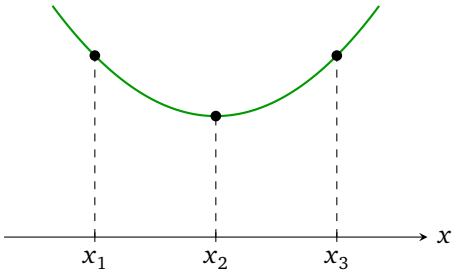
> **Version 1 of the argument:** Pick a zero of f, as long as there is one, let's call it $a_1 \in k$ and divide f by $(x - a_1)$ (in the sense of long division). It get's divided with the 0 remainder. So, we have $f = (x = a_1) \cdot g$ where g has degree $d - 1$ (the degree drops). If g is divisible by $(x - a_1)$, keep dividing. Once this process of dividing by $(x - a_1)$ is finished, we have $f = (x - a_1)^{p_1} \cdot h(x)$, where the degreee of h is smaller than the degree of g and a_1 is not a zero of h. If there is another zero, say $a_2 \in k$, do the same obtaining $f = (x - a1)^{p_1} \cdot (x - a_2)^{p_2} \cdot u(x)$, where a_1 and a_2 are both not zero of $u(x)$. Iterating this way, we end up at the representation $f = (x - a_1)^{p_1} \cdot (x - a_2)^{p_2} \cdot \cdots \cdot (x - a_m)^{p_m} \cdot w(x)$ where $a_1, \ldots, a_m \in k$, $degf = p_1 + \ldots + p_m + deg(w)$ and w has no zero in k. This shows that a non-zero polynomial of degree $d \in \mathbb{N}_0$ with at most $d \in \mathbb{N}_0$ zeros, where d is finite, but our field k is infinite. So, $f : k \to k$ is only zero on $\{a_1, \ldots, a_m\}$ and so it's not a zero function.

Theorem 1.1.8 (Continued)

Version 2 of the argument for $n = 2$: Let's use linear algebra and polynomial interpolation. Let's consider all polynomial of degree at most $d \in \mathbb{N}$, where d is fixed and arbitrary. Denote the set of these polynomials by $k[x]_d = \{\sum_{i=0}^{d} c_i x^i : c_i \in k \text{for every} i\}$. This is a k-vector space of dimension of $d+1$. We fix $d+1$ dstinct values $a_1, \ldots, a_{d+1}$ in k (our field is infinite, so no matter how large our d is, we always have such a choice) and consider the linear map that evaluates $f \in k[x]_d$ at these values.

$$ev(f) = (f(a_1), \ldots, f(a_{d+1}))$$

This is the map $ev : \underbrace{k[x]_d}_{\text{dimension } d+1} \to \underbrace{k^{d+1}}_{\text{dimension } d+1}$.



We're going to show that this map is surjective (in the language of the numerical mathematics: the polynomial interpolation task is solvable).

Let's fix an arbitrary $y = (y_1, \ldots, y_{d+1}) \in k^{d+1}$ and determine an $f \in k[x]_{d+1}$ satisfying $f(a_i) = y_i$ for all $i \in \{1, \ldots, d+1\}$.
Introduce

$$l_i(x) = \prod_{j=1\ldots,d+1: j \neq i} \frac{x - a_j}{a_i - a_j} \in k[x]_d$$

which satisfies

$$l_i(a_j) = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases}$$

Now, it is clear that $f = y_1 \cdot l_1 + \ldots y_{d+1} \cdot l_{d+1}$ satisfies $f(a_i) = y_i$ for all $i = 1, \ldots, d+1$. We have shown that $ev : k[x]_d \to k^{d+1}$ is surjective. So, from linear algebra, we know that f is also injective so that only the zero polynomial is mapped to $(0, \ldots, 0)$ by ev.

If you like more details on that point: by the rank-nullity theorem,

$$\underbrace{\dim \operatorname{im}(ev)}_{\text{dimension} d+1} + \underbrace{\dim \ker(ev)}_{\text{dimension} d+1} = \dim k[x]_d,$$

so that $\dim \ker(ev) = 0$.

We have done the case $n = 1$ and we continue with the inductive argument. Assume that the assertion is already verified for polynomials in $n-1$ variables for some $n \geq 2$. Based on this, we verify the assertion for polynomials in n variables.

We can write a polynomial $f \in k[x_1, \ldots, x_n]$ as an element of $k[x_1, \ldots, x_{n-1}][x_n]$, so we write $f = \sum_{i=1}^{d} g_i(x_1, \ldots, x_{n-1}) \cdot x_n^i$ with $g_1, \ldots, g_d k[x_1, \ldots, x_{n-1}]$. We fix arbitrary values for $x_1, \ldots, x_{n-1}$, which we denote by $a_1, \ldots, a_{n-1} \in k$. Fixing these values, we get $f(a_1, \ldots, a_{n-1}, x_n) = \sum_{i=1}^{d} g_i(a_1, \ldots, a_{n-1}) \cdot x_n^i \in k[x_n]$. By the $n = 1$ case of our argument, whenever $f(a_1, \ldots, a_{n-1}, x_n)$ defines the zero function $k \to k$, it is a zero polynomial. Hence, its coefficients $g_i(a_1, \ldots, a_{n-1})$ are a;; zero. But since $(a_1, \ldots, a_{n-1})$ are arbitrary values from k, the induction assumption implies that $g_i(x_1, \ldots, x_{n-1})$ is also a zero polynomial. Consequently, f is a zero polynomial.

□

Corollary 1.1.9

Let the field k be infinite. Then two polynomials $f, g \in k[x_1, \ldots, x_n]$ are equal if and only if there respective polynomial functions $f, g : k^n \to k$ are equal.

Proof: By 1.1.8 the difference function $f - g : k^N \to k$ is a zero function if and only if $f - g$ is a zero polynomial. This immediately gives the assertion.

□

Fun Fact: There is also a multivariatte polynomial interpolation, which is even more fun than the univariat and helps in this case.

Definition 1.1.10

A field k is called *algebraically closed* if for every polynomial $f \in k[x]$ of degree ≥ 1 there exists an $a \in k$ such that $f(a) = 0$.
(That is, each univariat polynomial equation over k of degree ≥ 1 has a solution in k.)

Theorem 1.1.11

The field $\mathbb{C}$ of complex numbers is algebraically closed.
(This result is called the fundamental theorem of algebra and we don't prove is here.)

Small Exercise 1.1.11a

Let k a finite field. Show that k is not algebraically closed.

Solution:

- **Leo:** $k = \{k_1, \ldots, k_n\}$ and $f = (k_1 - x) \cdot (k_2 - x) \cdots + k_1$, where $k_1 \neq 0$
- **Prof. Averkov:**

$$f = \prod_{s \in k} (x - s) + 1$$

$f(a) = 1$ for all $a \in k$ by construction. The degree of f is $deg f = |k| \geq 2 \geq 1$.

Add On: Let's take the field $\mathbb{Z} = \mathbb{F}_2$. Please write down this polynomial for this particular field. (Expand brackets)

Solution:

$$f = \prod_{s \in k} (x - s) + 1 = (x - 1) \cdot (x - 0) + 1 = (x + 1) \cdot x + 1 = x^2 + x + 1$$

This polynomial will help you to create another finite field.

1.2 Affine Varieties

Definition 1.2.1

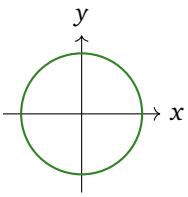
Let $f_1,\dots,f_s \in k[x_1,\dots,x_n]$ be finitely many polynomials ($s \in \mathbb{N}_0$). Then the set $V(f_1,\dots f_s) = \{(a_1,\dots,a_n) \in k^n : f_i(a_1,\dots,a_n) = 0 \forall i = 1,\dots,s\}$ is called the *affine variety* defined by the polynomials $f_1,\dots,f_s$. This is the solution set over the field k of the system of polynomial equations.

$$\begin{cases} f_1 = 0 \\ \quad \dots \\ f_s = 0 \end{cases}$$

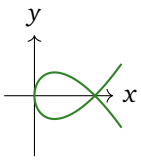
To indicate the underlying field explicitly, one also writes $V_k(f_1,\dots,f_s)$.

Example 1.2.2

1. $V_{\mathbb{R}}(x^2+y^2-1)$



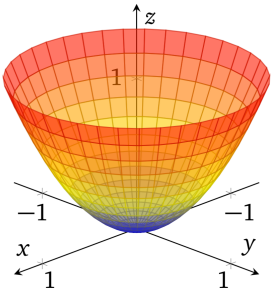
2. $V_{\mathbb{R}}(y^2-x(x-1)^2)$



Cubic elliptic curve over $\mathbb{R}$ with a singularity. In cryptography one uses elliptic curves over finite fields without singularities for encryption purposes.

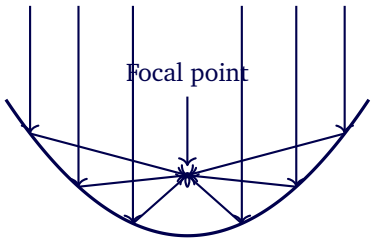
Whenever you use firefox you use elliptic curve encryption.

3. $V_{\mathbb{R}}(z-x^2-y^2)$



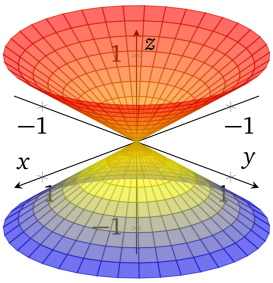
Paraboloid

***Side Note**: Why is a satelite dish parabolic?*



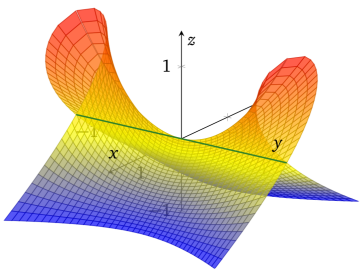
Car lights work the other way around with the light source in the focal point.

4. $V_{\mathbb{R}}(z^2-x^2-y^2)$



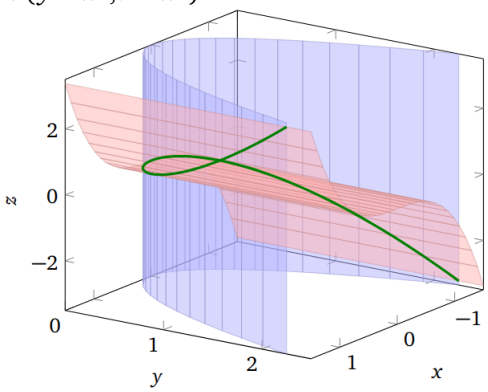
Cone

5. $V(x^2-y^2\cdot z^2+z^3)$



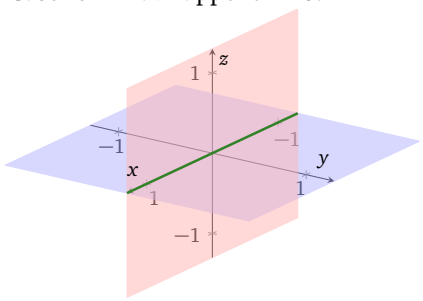
Example 1.2.2 continued

6. $V(y-x^2, z-x^3)$

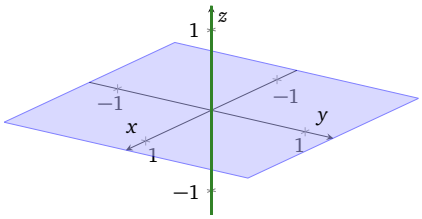


This is a curve, so called twisted cubic.

Linear Case for what happend in 6.



7. $V(xz, yz)$



This is a union of a line and a plane.

8. $V_{\mathbb{R}}(x^2+y^2+1) = \emptyset$ vs. $V_{\mathbb{C}}(x^2+y^2+1) \neq \emptyset$

9. $V(xy, xy-1) =$ for every field k.
Basic Task: How to figure out, if the variety is empty.

10. You can take an algebraic perspective of viewpoint on many topics on math. For example, optimization. Suppose you optimize $f(x,y,z)$ suspect to $g(x,y,z)=0$ where $f, g \in k[x,y,z]$.

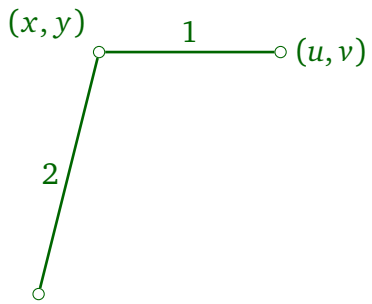
We can do Lagrange multipliers.

$$\begin{cases} \frac{\nabla f}{\nabla x} = \lambda \frac{\nabla g}{\nabla x} \\ \frac{\nabla f}{\nabla y} = \lambda \frac{\nabla g}{\nabla y} \\ \frac{\nabla f}{\nabla z} = \lambda \frac{\nabla g}{\nabla z} \end{cases}$$

$$\rightsquigarrow V(\frac{\nabla f}{\nabla x} - \lambda \frac{\nabla g}{\nabla x}, \frac{\nabla f}{\nabla y} - \lambda \frac{\nabla g}{\nabla y}, \frac{\nabla f}{\nabla z} - \lambda \frac{\nabla g}{\nabla z})$$

We work in the ring $k[x,y,z,\lambda]$.

11. $V(x^2+y^2-4, (x-u)^2+(y-v)^2-1)$



Two-dimensional variety in the space of dimension four.

12. Solution set of linear systems of equations (these are called *affine subspaces*).

20.10.2026

(Fix the definition of late to one minute.)

Lemma 1.2.3

Let $V, W \subseteq k^n$ be affine varieties. Then $V \cap W$ and $V \cup W$ are also affine varieties.

Proof: Let $V = V(f_1, \ldots, f_l)$ and $W = V(g_1, \ldots, g_m)$. For finitely many polynomials $f_1, \ldots, f_l, g_1, \ldots, g_l \in k[x_1, \ldots, x_n]$. The $V \cap W = V(f_1, \ldots, f_l, g_1, \ldots, g_m)$, so $V \cap W$ is a variety.
It turns out that $V \cup W = V(f_i \cdot g_j : i = 1, \ldots, l, j = 1, \ldots, m)$. Let's varify this. Let $a = (a_1, \ldots, a_n) \in V \cup W$. This means $a \in V$ or $a \in W$. If $a \in V$, then $f_1(a) = f_2(a) = \ldots = f_i(a) = 0$. So, $f_i(a) \cdot g_j(a) = 0$ for all i and j. $\Rightarrow a \in V(f_i \cdot g_j : i = 1, \ldots, l, j = 1, \ldots, m)$.
For $a \in W$, the argument is analogous.
To conclude the proof, we consider $a \notin V \cup W$ and show that $a \in V(f_i \cdot g_j : i = 1, \ldots, l, j = 1, \ldots, m)$. Since, $a \notin V \cup W$, we have $a \notin V$ and $a \notin W$. Since, $a \notin V$, we have $f_s(a) \neq 0$ for some $s = 1, \ldots, l$. Since, $a \notin W$, we have $g_t(a) \neq 0$ for some $t = 1, \ldots, m$. $\Rightarrow$ $f_s(a) \cdot g_t(a) \neq 0$ $\Rightarrow$ $a \notin V(f_i \cdot g_j : i = 1, \ldots, l, j = 1, \ldots, m)$ □

Remark 1.2.3a

Any finite subset of k^n is variety. These are so-called zero-dimensional varieties.

Remark 1.2.3b

Now, you can understand $\underbrace{V(xz, yz)}_{\text{decomposable}} = V(x, y) \cup V(z)$ $\rightsquigarrow$ irreducible varieties.

Remark 1.2.3c

When we were in the kindergarten, we learn linear algebra; the kid whose name was Gauss (smart kid with no friends) told us how to solve things.

(Now we are not in kindergarten anymore, we need to deal with hard life.)

We can ask a few questions about a variety $V = V(f_1, \ldots, f_l)$ for given polynomials $f_1, \ldots, f_l \in k[x_1, \ldots, x_n]$.

1. Is $v \neq 0$, i.e. does the system $f_1 = \ldots = f_l = 0$ have a solution over k?
2. Is V finite?
3. What is the dimension of V? (We still need to define dimension and it's not that easy)

The hardness of these problems depends on the field.

Exercise 1.2.4

a) For the fields you know, decide if they are algebraically closed.
b) For subsets of k^n that you know, decide if they are affine varieties.
Examples: $\mathbb{R}_{\geq 0}$ - is it a variety? , $k \backslash \{0\}$ - is it a variety?

1.3 An Example of a Topic that involves Varieties and Polynomials: Parametrization and Implicitization

Example 1.3.1 (Linear Case)

Let's consider a line, given implicitly, that means by equations.

$$\begin{cases} x+y+z=1 \\ x+2y-z=3 \end{cases}$$

Our field is $\mathbb{R}$ for this example. Describe this line parametically. Let's call the parameter t.

(Flashback to Kindergarten!!)

Solution: → same setting as linear case in Example 1.2.2 (6.)

$$\begin{cases} x+y+z=1 \\ x+2y-z=3 \end{cases} \Longrightarrow \begin{cases} x+y+z=1 \\ y-2z=-2 \end{cases} \Longrightarrow \begin{cases} x+3z=-1 \\ y-2z=2 \end{cases}$$

Solutionset:

$$\{(x,y,z) : x=-1-3t, y=2+2t, z=t, t\in\mathbb{R}\}$$

Remark 1.3.2

The linear case is done by linear algebra: one can parameterize the solution set of a system of linear equations via Gaussian elimination.

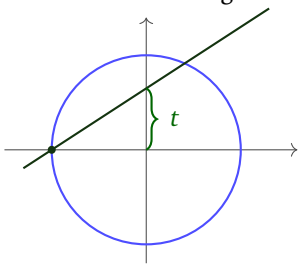
(Go through the details yourself or otherwise ask.)

Example 1.3.3 (Quadratic Case)

Let's consider everyone's favorite: the circle $x^2+y^2=1$.
The circle has a rational parametrization:

$$x=\frac{u(t)}{g(t)}, y=\frac{v(t)}{g(t)}, u,v,g\in\mathbb{R}[t]$$

Now consider the following:



The slope of the line can be changed, which is represented by the parameter t. Find a representation of the other intersection of the line with the circle (in rational parameterization).

26.10.2026

Solution via Breathing-Procedure:
First thing we want is an equation of the line:

$$y=t\cdot(x+1)$$

Combining both equations, we get:

$$x^2+(t\cdot(x+1))^2=1$$

We know the term $(x-1)$ and have the following factorization:

$$(x+1)\cdot(\ldots x+\ldots)=0$$

Example 1.3.3 (Quadratic Case) Continued

Now determine the other factor:

$$(x+1)\cdot((1+t^2)\cdot x - 1 + t^2) = 0$$

For $x \neq 0$, we have

$$x = \frac{1-t^2}{1+t^2} \text{ and } y = t(x+1) = t\left(\frac{1-t^2}{1+t^2}+1\right) = \frac{2t}{1+t^2}$$

Double check:

$$\begin{aligned} x^2+y^2 &= \frac{(1-t^2)^2}{(1+t^2)^2} + \frac{4t^2}{(1+t^2)^2} \\ &= \frac{1-2t^2+t^4+4t^2}{(1+t^2)^2} \\ &= \frac{1+2t^2+t^4}{(1+t^2)^2} \\ &= 1 \end{aligned}$$

Definition 1.3.4 (Rational Functions)

A rational function with coefficients in the field k in the variables $x_1, \ldots, x_n$ is a formal quotient $\frac{f}{g}$ where $f, g \in k[x_1, \ldots, x_n]$ and $g \neq 0$. The set of all rational functions is denoted by $k(x_1, \ldots, x_n)$ (round brackets).

Two rational functions $\frac{f}{g}$ and $\frac{a}{b}$ are defined to be equal when $f \cdot b = a \cdot g$.

Rational functions can be added and multiplied as follows:

$$\frac{f}{g} + \frac{u}{v} := \frac{f\cdot v + g\cdot u}{g\cdot v} \quad \text{and} \quad \frac{f}{g}\cdot\frac{u}{v} := \frac{f\cdot u}{g\cdot v}$$

for $f, u, g, v \in k[x_1, \ldots, x_n], g \neq 0, v \neq 0 \ldots$

The above definitions are well-defined, because in the ring $k[x_1, \ldots, x_n]$:

$$g \neq 0 \text{ and } v \neq 0 \Rightarrow g \cdot v \neq 0$$

(This needs to be shown and it is not true in every ring.)

Remark 1.3.5

$k[x_1, \ldots, x_n]$ is a field (which is great!).

Definition 1.3.6

Let $V \subseteq k^n$ be a variety. We say that the expressions

$$x_1 = \frac{p_1(t_1, \ldots, t_m)}{q(t_1, \ldots, t_m)}, \ldots, x_n = \frac{p_n(t_1, \ldots, t_m)}{q(t_1, \ldots, t_m)}$$

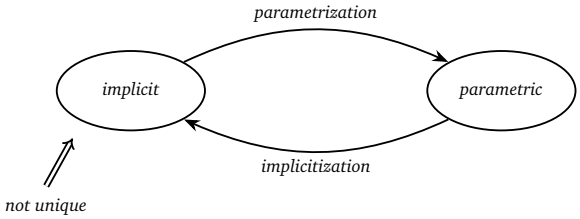
where $p_1, \ldots, p_n, q \in k[x_1, \ldots, x_n]$ and $q \neq 0$ parameterize the variety V if V is the smallest variety w.r.t unclusion that contains the set

$$\left\{\left(\frac{p_1(t_1, \ldots, t_m)}{q(t_1, \ldots, t_m)}, \ldots, \frac{p_n(t_1, \ldots, t_m)}{q(t_1, \ldots, t_m)}\right) : a_1, \ldots, a_m \in k, q(a_1, \ldots, a_m) \neq 0\right\}$$

A variety that has such a parameterization (which is called rational) is called a *rational variety*.

Definition 1.3.7

The description of a variety V by a system of finitely many equations $f_1 = \ldots = f_m = 0$ where $f_1, \ldots, f_m \in k[x_1, \ldots, x_n]$ is called an *implicit description* (this is our default description, it always exists by the definition of a variety).

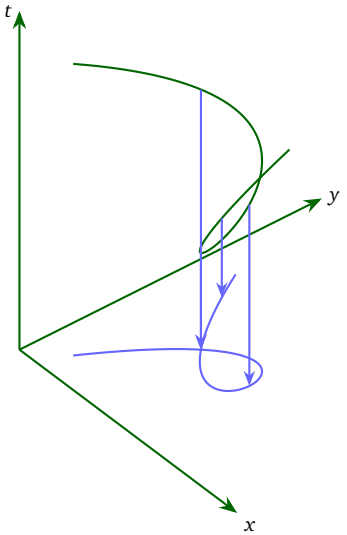


Example 1.3.8

Bezier splines is a cursive given parameterically, whose x and y coordinates are parameterized as cubic polynomials. Normally written like this:

$$\begin{bmatrix} x \\ y \end{bmatrix} = (1-t)^3 P_0 + 3(1-t)^2 t P_1 + 3(1-t) t^2 P_2 + t^3 P_3$$

where $P_0, \ldots, P_3 \in k^2$.
One may assk for an implicit equation of such a curve.



Making things implicit $\rightsquigarrow$ projection

Remark 1.3.9

In k^n one can define an affine subspace by a linear system as $\{x : Ax = b\}$ where $Ax = b$ is a linear system over k and parametrically as $\{U \cdot t + v : t \in k^m\}$, where U and v are a matrix reps. a vector over k. So, we have a linear version of our tasks.
Both tasks are solvable with Gaussian Elimination.

Exercise 1.3.10

1. Why is the product of two non-zero polynomials in $k[x_1, \ldots, x_n]$ a non-zero polynomial?

2. Find an implicit description of the parametric curve $x = \frac{t}{t+t^2}$ and $y = 1 - \frac{1}{t^2}$.

3. Justify Remark 1.3.9.

4. Find a rational representation of the sphere $V(x^2 + y^2 + z^2 - 1)$.

5. Find a parametric description of the variety $V(y^2 - x^2 + x^3)$.

6. Implicitize

$$\begin{aligned} x_1 &= t_1 + t_2 \\ x_2 &= t_1 + 2t_2 \\ x_3 &= 2t_1 + t_2 \\ x_4 &= t_1 - t_2 \end{aligned}$$

   Where t_1, t_2 are parameters.

The field is $k = \mathbb{R}$.

1.4 Ideals

27.10.2026

A key notion in algebra of the rings.

Definition 1.4.1

Let $(R,+,\cdot)$ be a commutative ring. Then a subset I of R is called an *ideal* of R if the following holds:
(a) $0 \in I$
(b) For all $f,g \in I$, one has $f+g \in I$
(c) For all $f \in R$ and $g \in I$, one has $f \cdot g \in I$

The definition looks a little like the notion of a vector subspace.

You can view I as a set of objects that are zero in some abstract sense.

One can define ideals by system of generators.

Definition 1.4.2

Let R be a commutative ring and $f_1,\dots,f_s$ finitely many elements of R. We define the ideal generated by $f_1,\dots,f_s$ as follows:

$$\langle f_1,\dots,f_s\rangle_R = \{f_1\cdot g_1+\dots+f_s\cdot g_s \mid g_1,\dots,g_s \in R\}$$

If the choice of R is clear the subscript R can be omitted.

Lemma 1.4.3

For $f_1,\dots,f_s$ in a commutative ring R, the set $\langle f_1,\dots,f_s\rangle_R$ is an ideal.

Proof: try poving this. Not really hard.

Example 1.4.4 (The use of ideals in the context of implicitization)

Let's consider an easy parametric curve given by

$$x=t,\quad y=t^2,\quad z=t^3.$$

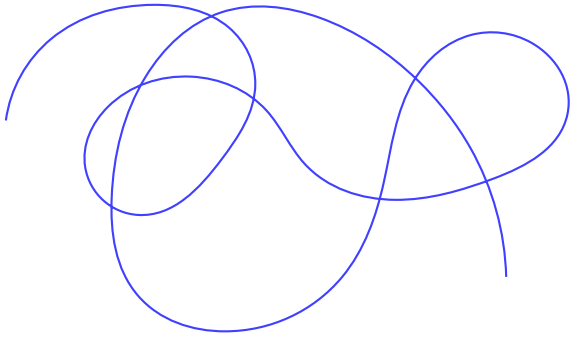
where t is a parameter. This is a space curve. Let's write those equations as

$$x-t=0,\quad y-t^2=0,\quad z-t^3=0,$$

where $x-t, y-t^2, z-t^3 \in k[x_1,\dots,x_n]$. The ideal $I=\langle x-t, y-t^2, z-t^3\rangle$ contains polynomials $f(x,y,z,t)$ that define equations $f(x,y,z,t)=0$ deriviable out of our thee equations (derivable in the sense of 1.4.2).
It turns out that some polynomials in I don't depent on t.

$$\left.\begin{aligned}(x+t)\cdot(x-t)+(-1)\cdot(y-t^2)&=x^2-y\\ (x^2+xt+t^2)\cdot(x-t)+(-1)\cdot(z-t^3)&=x^3-z\end{aligned}\right\} \leftarrow \text{gives two implicit equations for our curve.}$$

In more complicated situations, you won't see the implicit equations to easily.



$$x=t-t^2+3t^3-t^{10}$$
$$y=t+t^5-t^7$$

Example 1.4.4 Continued

There is a maschinery that helps us to find implicit equations in such cases. It is called *Gröbner basis*. We can use *SageMath* to solve such problems.

SageMath Code

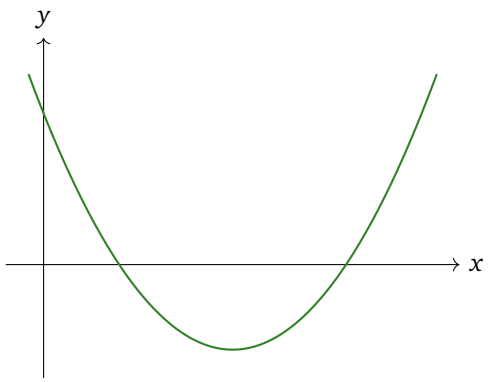
```
R = PolynomialRing(QQ, 'x,y,t')
x,y,t = R.gens()
I = R.ideal(x -(t - t^2 + 3*t^3 - t^10), y - (t + t^5 - t^7))
I.elimination_ideal(t) <-- done with Gröbner bases
```

Let $u(t), v(t) \in kt$ and a parametric curve given by

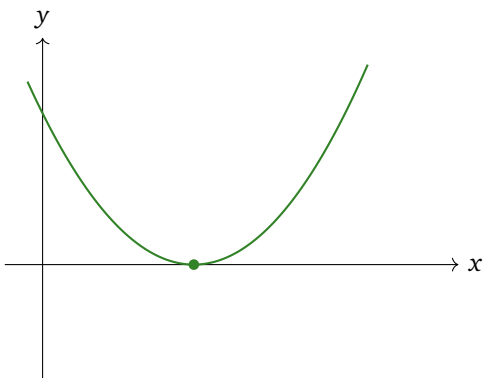
$$x = u(t), \quad y = v(t)$$

Introduce the ideal $\langle x - u(t), g - u(t)\rangle \subseteq k[x, y, t]$, one has the so-called *eliminaton ideal* $I \cap k[x, y]$. The task is to generate this ideal.

Example 1.4.4a



Discriminant $\neq 1$



Discriminant $= 0$
Derivative $= 0$

$$I = \langle x^2 + px + q, 2x + p\rangle \subseteq k[x, p, q]$$

$I \cap k[p, q] \leftarrow$ the generator will essentially be the discriminant

Definition 1.4.5

An ideal I in a commutative ring R is called *finitely generated* if $I = \langle f_1, \ldots, f_s\rangle$ for some finitely many $f_1, \ldots, f_s \in R$. In this case $\{f_1, \ldots, f_s\}$ is called a *set of generators* of I.

Key Question: Are ideals in the $k[x_1, \ldots, x_n]$ finitely generated?

Now, we have an interplay: ideals vs. varieties.

$$V(f_1, \ldots, f_s) \qquad \langle f_1, \ldots, f_s\rangle$$

One can view the variety $V(f_1, \ldots, f_s)$ not as a function of $(f_1, \ldots, f_s)$ but as a function og the ideal $\langle f_1, \ldots, f_s,\rangle$.

Proposition 1.4.6

$$\langle f_1, \ldots, f_s \rangle = \langle g_1, \ldots, g_s \rangle$$

Then

$$V(f_1, \ldots, f_s) = V(g_1, \ldots, g_s).$$

Proof: You'll derive it by spelling out the underlying notation. Try!

Definition 1.4.7

For a set $X \subseteq k^n$ we introduce the *ideal of* X by:

$$I(X) = \{f \in k[x_1, \ldots, x_n] \mid f(a) = 0 \text{ for all } a \in X\}$$

Lemma 1.4.8

For $X \subseteq k^n$, the set $I(X)$ is in fact an ideal.

Proof: Just spell out $I(X)$ and use the definition of an ideal. Try!

Example 1.4.9

1. $V = \{(0,0)\}$ variety in k^2.

$$I(V) = \{f \in k[x,y] \mid f(0,0) = 0\} = \langle x, y \rangle$$

Polynomials in the two variabeles:

$$\sum_{i,j} c_{ij} x^i y^j \quad \Rightarrow f(0,0) = c_{0,0}$$

02. and 03. 11.2026

Short Recap

$$\text{Variety} \quad \Leftrightarrow \quad \text{Ideal}$$

$$X \subseteq k^n \quad \mapsto \quad I(X) = \{f \in k[x_1, \ldots, x_n] : f \text{ vanishes on } X\}$$

$$V(f_1, \ldots, f_m) = \{a \in k^n : f_1(a) = \ldots = f_m(a) = 0\}$$

$$\langle f_1, \ldots, f_m \rangle \quad \mapsto \quad V(f_1, \ldots, f_m) \subseteq k^n \quad \mapsto \quad \underbrace{I(V(f_1, \ldots, f_m))}$$

The set of polynomials, that vanish whenever the polynomials $f_1, \ldots, f_m$ vanish.

One would frequently study a variety $V \subseteq k^n$ by studying the respective ideal $I(V)$.

Lemma 1.4.10

For $f_1, \ldots, f_s \in k[x_1, \ldots, x_n]$, one has $\langle f_1, \ldots, f_s \rangle \subseteq I(V(f_1, \ldots, f_s))$ Moreover, there exist choices of $f_1, \ldots, f_s$ where whis inclusion is strict.

Proof:

$$g \in \langle f_1, \ldots, f_s \rangle \Leftrightarrow g = f_1 \cdot h_1 + \ldots + f_s \cdot h_s$$

for some $h_1, \ldots, h_s \in k[x_1, \ldots, x_n]$

Lemma 1.4.10 continued

$$g \in I(V(f_1,\dots,f_s)) \Leftrightarrow g \text{ vanishes on } V(f_1,\dots,f_s)$$
$$\Leftrightarrow g(a) = 0 \text{ whenever } f_1(a) = \dots = f_s(a) = 0, \text{ for } a \in k^n.$$

This explains the inclusion in the assertion. For strict inclusions, consider $k[x]$ with $V(x^2)$:
- $\langle x^2 \rangle$ = set of all polynomials that do not have constant and linear terms.
- $V(x^2) = \{0\} \quad \Rightarrow \quad I(V(x^)) =$ set of polynomials that do not have the constant term.

Consequently, $x \in I(V(x^2))$ but $x \in \langle x^2 \rangle$.

□

The so-called *Nullstellensatz* clarifies the relation of $I(V(f_1,\dots,f_s))$ to $\langle f_1,\dots,f_s\rangle$ in the algebraic term in the case when the field k is algebraically closed (for example $k = \mathbb{C}$).

Proposition 1.4.11

Let $V, W \subseteq k^n$ be affine Varieties. Then the following hold:
(i) $V \subseteq W \Rightarrow I(V) \supseteq I(W)$.
(ii) $V = W \Leftrightarrow I(V) = I(W)$.

Proof:
(i) follows direktly from our definition of $I(X)$ for a set $X \in k^n$.
(ii) $V = W \quad \Rightarrow I(V) = I(W)$ is trivial.

Conversely, assume $I(V) = I(W)$. Let $V = V(f_1,\dots,f_s)$ and $W = V(g_1,\dots,g_t)$ mit $f_1,\dots,f_s,g_1,\dots,g_t \in k[x_1,\dots,x_n]$.
Equality $I(V) = I(W)$ means that every polynomial vanishing on V also vanishes on W and vice versa.

$$f_1,\dots,f_s \text{ vanishes on} V \Leftrightarrow g_1,\dots,g_t \text{ vanishes on} W$$

So, $W \subseteq V(f_1,\dots,f_s) = V$. Exchanging the roles of V and W, we also have the other inlcusion $V \subseteq W$.

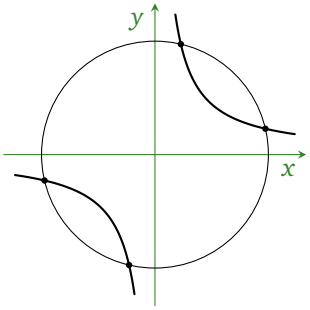
$$\Rightarrow V = W$$

□

$$\begin{array}{ccc} f_1 = \dots = f_s = 0 & \longrightarrow & \langle f_1,\dots,f_s\rangle \\ \downarrow & & \supseteq \\ V(f_1,\dots,f_s) & \longrightarrow & I\big(V(f_1,\dots,f_s)\big) \end{array}$$

Nullstellensatz (für k algebraisch abgeschlossen)

Exercise 1.4.12



1. Consider the drawing on the left.

$$\begin{cases} x^2 + y^2 - 1 = 0 \\ 2xy - 1 = 0 \end{cases}$$

Find a polynomial $f(x) \in k[x]\backslash\{0\}$ such that the equation $f(x) = 0$ follows from this system.
Is $f \in \langle x^2 + y^2 - 1, 2xy - 1\rangle$?
2. $k = \mathbb{F}_2, K^2$. What is $I(k^2)$? Describe $I(k^2)$ by a set of generators.

1.5 Polynomials in one variable

Definition 1.5.1

For $f = a_o + a_1 x + \ldots + a_m x^m \in k[x]\backslash\{0\}$ with $a_0, \ldots, a_m \in k$ and $a_m \neq 0$, we call $a_m x^m$ the *leading term of* f.
Notation: $LT(f)$

Proposition 1.5.2

For every choice of $f \in k[x]$ and $g \in k[x]\backslash\{0\}$ there exisyts a unique decomposition

$$f = q \cdot g + r$$

with $q, r \in k[x]$ and $\deg r < \deg g$. Furthermore, there exists an algrithm that calculates g and r for given f and g.

Proof: In our algorithm we start with

$$\underbrace{q = 0}_{\text{quotient in progress}} \quad \text{and} \quad \underbrace{r = f}_{\text{remainder in progress}}$$

We will maintain the equality $f = g \cdot q + r$ as the invariant, making the $\deg r$ smaller in each iteration until the condition $\deg r < \deg g$ is satisfied.
If $\deg r < \deg g$, then we can terminate. Otherwise, we want to update r using g to make r smaller.

$$\deg r \geq \deg g \implies f = g \cdot \underbrace{q + \frac{LT(r)}{LT(g)}}_{\text{new } q} + \underbrace{r - \frac{LT(r)}{LT(g)} \cdot g}_{\text{new } r}$$

by taking into account that $f = g \cdot q + r$.
This leads to:

Univariate Division Algorithm

Assumption: $f, g \in k[x], g \neq 0$
Result: quotient q and the remainder r of the long division of f by g.

```
r:=f    % Remainder in progress
q:=0    % Quotient in progress

while deg r >= deg g:
    T := LT(r)/LT(g)
    q := q + T   % update of the quotient in progress
    r := r - T * g  % update of the remainder in progress

return (q,r)
```

The algorithm is correct and is terminating in finitely many steps, because we maintain the invariant $f = g \cdot q + r$ and the updates of q and r are such that $\deg r$ gets strickly smaller with each iteration.

It remains to show the uniqueness of the decomposition.
Assume $f = q \cdot g + r = g \cdot Q + R$ with $q, r, Q, R \in k[x]$ and $\deg r < \deg g$ and $\deg R < \deg g$.
If $q = Q$, then $r = f - g \cdot q = f - g \cdot Q = R$ and so $r = R$.
If $q \neq Q$, then $0 = f - f = (g \cdot Q - R) - (g \cdot q - r) = g \cdot (Q - q) + r - R$

$$\underbrace{R - r}_{degree < \deg g} = \underbrace{g}_{\deg g} \cdot \underbrace{(Q - q)}_{\neq 0}$$

The left-hand side has degree $< \deg g$, but the right-hand side has degree $\geq \deg g$, which is a contradiction.

$$\implies Q = q$$

□